



ST. CATHERINE'S SCHOOL

YEAR 12 TRIAL EXAMINATION

3/4 UNIT MATHEMATICS

TIME ALLOWED: 2 HOURS (PLUS 5 MINUTES READING TIME)

DATE: AUGUST 1999

STUDENT NUMBER: _____

DIRECTIONS TO CANDIDATES:

- This paper consists of seven questions.
- All questions are to be attempted.
- All questions are of equal value.
- In every question, all necessary working should be shown.
- Marks may be deducted for careless or badly arranged work.
- Approved calculators and geometrical instruments are required.
- Begin a NEW PAGE for every question.
- Attach your question paper to the front of Section A.
- Hand your work in three bundles:
 - Section A - Questions 1, 2 and 3
 - Section B - Questions 4, 5, 6 and 7
- This sheet will form the cover page for Section A. You will need to write a cover sheet for Section B, which clearly states your Student Number.

Securely staple or tie questions together in sections.

TEACHERS USE ONLY TOTAL MARKS	
A	
B	

3 Unit Trial Mathematics Examination Paper 1999

Section A

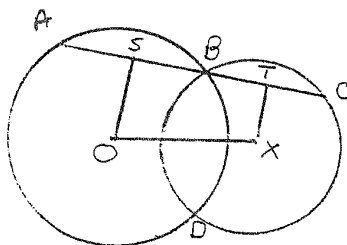
Question 1

Marks

- a) Solve the inequality $\frac{x^2 - 1}{x} > 0$ 3
- b) Evaluate $\int_0^{\pi} \sin^2 x \, dx$ 3
- c) Integrate $\int \frac{t}{\sqrt{1+t}} \, dt$ by using the substitution $t = u^2 - 1$ 3
- d) A particle moves from rest from the origin in a straight line in such a way that its velocity $v \, m/s$ is given by $v = 20t - 5t^2$, (where t is in seconds). 3
- Find (i) when the particle comes to rest
- (ii) the greatest velocity of the particle.

Question 2 (Start a new page)

- a) If $A(x)$ is a factor of $P(x)$, find a when $A(x) = x - 4$ and $P(x) = x^3 + 2x^2 + ax - 20$ 2
- b) Express $12\cos\theta + 5\sin\theta$ in the form $R\cos(\theta - \alpha)$ and use it to solve $12\cos\theta + 5\sin\theta = 13$ for $0^\circ \leq \theta \leq 360^\circ$. 5
- c) The equation $e^x = x + 2$ has a root close to $x = 1.2$. Use Newton's method once to find a better approximation to this root (correct to 2 decimal places). 2
- d) ABC is a straight line
S and T are midpoints of AB and BC respectively
O is centre of circle ABD
X is centre of circle BCD 3



Prove $\angle SOX$ is the supplement of $\angle OXT$.

Question 3 (Start a new page)

Marks

a) Consider the function $f(x) = \frac{x}{x^2 + 1}$

7

(i) Show that it is an odd function

(ii) Find any stationary points and given that

$f''(x) = \frac{2x(x^2 - 3)}{(x^2 + 1)^3}$, find any points of inflexion.

(iii) Describe the behaviour of $f(x)$ for very large positive and very large negative values of x
i.e. when $x \rightarrow \infty$ and $x \rightarrow -\infty$.

(iv) Sketch the curve.

b) Prove by mathematical induction that

$\sum_{r=1}^n r(r+2) = \frac{1}{6}n(n+1)(2n+7)$ where n is a positive integer.

5

Question 4

Question 4

3

- 3

3

3

Question 5 (Start a new page)

- 4

8

Question 6 (Start a new page)

Marks

- a) Find the co-ordinates of the point P which divides the interval AB with end points $A(2,3)$ and $B(5,-7)$ internally in the ratio 4:9. 2
- b) A sphere is expanding such that its surface area is increasing at the rate of $0.01 \text{ cm} / \text{sec}^2$. Calculate the rate of change of 5
- (i) its radius
(ii) its volume
- at an instant when the radius is 5 cm.
- c) Find $\frac{d}{dx} \sin^{-1} e^{2x}$ and hence evaluate $\int_{-\ln \sqrt{2}}^0 \frac{2e^{2x}}{\sqrt{1-e^{4x}}} dx$ 5

Question 7 (Start a new page)

Marks
6

- a) Brine, containing 1 kg of salt per 10 litres, runs into a tank, initially filled with 500 litres of fresh water, at a rate of 25 litres per minute.

The mixture runs out of the tank at the same rate of 25L/min.

- (i) If A is the amount of salt in the tank at time t , by calculating the concentration of salt flowing in and out of the tank,

show that $\frac{dA}{dt} = -\frac{1}{20}(A - 50)$

NOTE: 1 L of water weighs 1kg.

- (ii) Find the amount of salt in the tank at the end of 100 minutes, assuming that the mixture is kept uniform by stirring.

- b) A particle moves with an acceleration which varies linearly as the distance travelled such that $\ddot{x} = mx + b$. It starts at the origin from rest with an acceleration of $3m/s^2$ and reaches maximum speed in a distance of $160m$.

6

Find (i) the maximum speed

(ii) the speed when the particle has moved $80m$.

END OF EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

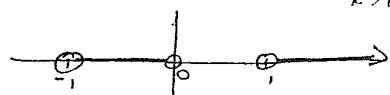
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

1a) $\frac{x^2-1}{x} > 0$

Consider $\frac{x^2-1}{x} = 0$

$x \neq 0$ $x^2-1=0$
 $x = \pm 1$



b) $\cos 2x = \cos^2 x - \sin^2 x$
 $= 1 - 2\sin^2 x$
 $\therefore \sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$

$\int_0^{\pi} \sin^2 x \, dx$
 $= \int_0^{\pi} \left(\frac{1}{2} - \frac{1}{2} \cos 2x \right) dx$
 $= \left[\frac{1}{2}x - \frac{1}{4} \sin 2x \right]_0^{\pi}$
 $= \frac{\pi}{2} - \frac{1}{4} \sin 2\pi - 0 + \frac{1}{4} \sin 0$
 $= \frac{\pi}{2} - \frac{1}{4} \times 0$
 $= \frac{\pi}{2}$

1) $\int \frac{t}{\sqrt{1+t}} dt$ where $t = u^2 - 1$
 $= \int \frac{u^2 - 1}{\sqrt{1+u^2-1}} \times 2u \, du$
 $= \int \frac{u^2 - 1}{u} \times 2u \, du$

$= 2 \left[\frac{u^3}{3} - u \right] + C$ $u^2 = t+1$
 $= 2 \frac{(t+1)^{3/2}}{3} - 2(t+1)^{1/2} + C$ $u = \sqrt{t+1}$

$t=0$ $v = 20t - 5t^2$
 $v=0$
 $x=0$

(i) rest when $v=0$
 $20t - 5t^2 = 0$
 $5t(4-t) = 0$
 $t=0$ $t=4$

(ii) greatest velocity

when $\frac{dv}{dt} = 0$ $\frac{dv}{dt} = 20 - 10t$
 $\frac{dv}{dt} = 0$ when $20 - 10t = 0$
 $t = 2$

St Catherine's
Trial HSC
Solutions 1999

2a) $A(x) = x - 4$ is a factor of $P(x)$

$\therefore P(4) = 0$

$P(x) = x^3 + 2x^2 + ax - 20$

$P(4) = 4^3 + 2(4)^2 + 4a - 20 = 0$

$76 + 4a = 0$
 $4a = -76$
 $a = -19$

b) $12 \cos \theta + 5 \sin \theta$

$= 13 \left(\frac{12}{13} \cos \theta + \frac{5}{13} \sin \theta \right)$

$= 13 (\cos \theta \cos \alpha + \sin \theta \sin \alpha)$

$= 13 \cos(\theta - \alpha)$

$= 13 \cos(\theta - 22.37^\circ)$

$\therefore 12 \cos \theta + 5 \sin \theta = 13$

$13 \cos(\theta - 22.37^\circ) = 13$

$\cos(\theta - 22.37^\circ) = 1$

$\therefore \theta - 22.37^\circ = -360^\circ, 0^\circ, 360^\circ, \dots$

$\theta = 22.37^\circ$ for $0 \leq \theta \leq 360^\circ$

c) $e^x = x + 2$

$\therefore e^x - x - 2 = 0$

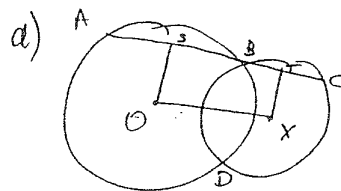
$P(x) = e^x - x - 2$

$P'(x) = e^x - 1$

$P(1.2) = e^{1.2} - 1.2 - 2 = 0.1201$

$P'(1.2) = e^{1.2} - 1 = 2.3201$

2nd approx $x_2 = 1.2 - \frac{P(1.2)}{P'(1.2)}$
 $= 1.15$



S is midpt of AB data

$\angle OSA = \angle OSB = 90^\circ$ join of centre to midpt of chord is perp to chord

Similarly T is midpt of BC data

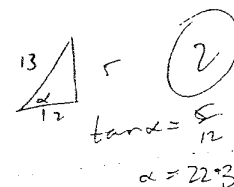
$\angle XTB = \angle XTC = 90^\circ$ join of centre etc

how $\angle OSB$ & $\angle XTB$ are co-interior and add to 180°
 $\therefore OS \parallel XT$

page 1

(2)

(16)



(1,0)

(3)

(2)

(3)

3 a) $f(x) = \frac{2x}{x^4+1}$

(i) $f(-x) = \frac{-2x}{(-x)^4+1}$
 $= \frac{-x}{x^4+1}$
 $= -\left(\frac{x}{x^4+1}\right)$
 $= -f(x)$

$\therefore f(x)$ is odd fn

(ii) $f'(x) = \frac{x^4+1(1) - x(2x)}{(x^4+1)^2}$
 $= \frac{1-2x^2}{(x^4+1)^2}$

St pt occur when $f'(x)=0$ i.e. $\frac{1-2x^2}{(x^4+1)^2} = 0$
 $x^2 = 1$
 $x = \pm 1$

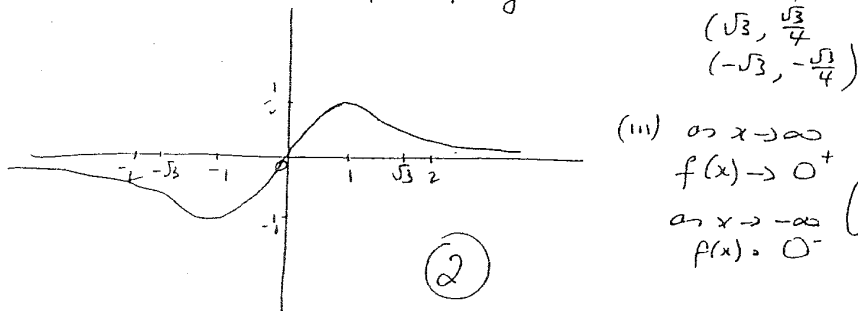
St pts $(1, \frac{1}{2})$ & $(-1, -\frac{1}{2})$

Pts of inflex may occur when $f''(x) = 0$
 $\frac{2x(x^4-3)}{(x^4+1)^3} = 0$

i.e. $x=0$ or $x = \pm\sqrt{3}$

check for concavity
 $x=-2$ $f''(-2) < 0 \wedge$
 $x=-1$ $f''(-1) > 0 \cup$
 $x=1$ $f''(1) < 0 \wedge$
 $x=2$ $f''(2) > 0 \cup$

as there are changes of concavity
 pts of inflex are at $(0,0)$



(iii) as $x \rightarrow \infty$
 $f(x) \rightarrow 0^+$
 as $x \rightarrow -\infty$
 $f(x) \rightarrow 0^-$

3b) $\sum_{r=1}^n r(r+2) = \frac{1}{6} n(n+1)(2n+7)$

range $\leftarrow$

1. Prove for $n=1$

LHS $= 1(1+2) = 3$ RHS $= \frac{1}{6}(1)(1+1)(2+7) = 3$

2. Assume true for $n=k$

i.e. $\sum_{r=1}^k r(r+2) = \frac{1}{6} k(k+1)(2k+7)$

3. Prove true for $n=k+1$

i.e. prove $\sum_{r=1}^{k+1} r(r+2) = \frac{1}{6} (k+1)(k+2)(2k+9)$

Proof: LHS $= \sum_{r=1}^{k+1} r(r+2)$

$= \sum_{r=1}^k r(r+2) + (k+1)(k+3)$

$= \frac{1}{6} k(k+1)(2k+7) + (k+1)(k+3)$

$= (k+1) \left[\frac{1}{6} (k(2k+7) + 6(k+3)) \right]$

$= \frac{1}{6} (k+1) (2k^2 + 7k + 6k + 18)$

$= \frac{1}{6} (k+1) (2k^2 + 13k + 18)$

$= \frac{1}{6} (k+1) (k+2)(2k+9)$

$= \text{RHS}$

4.

As it is true for $n=1$

then by step 3 it is true for $n=2$

As it is true for $n=2$

then it is true for $n=3$ and so on

$\therefore$ therefore, by the principle of mathematical induction

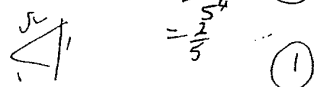
$\sum_{r=1}^n r(r+2) = \frac{1}{6} n(n+1)(2n+7)$ (must say this for $\forall n$)

2. a) (1) 2, 3, 4, 5, 6

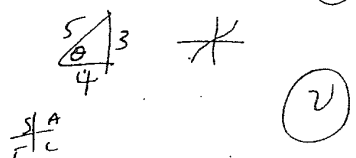
$\therefore 4^3 \times \boxed{} \boxed{} \boxed{} \boxed{2} = 2 \times {}^4P_3 = 48$ (1)

i) $\boxed{} \boxed{} \boxed{} \boxed{2} = 5^3 \times 2 \therefore P(\text{odd}) = \frac{5^3 \times 2}{5^4} = \frac{2}{5}$ (2) 5 a)

i) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$



(ii) $\cos(\sin^{-1}(\frac{3}{5}))$
 $= \cos(\theta)$
 $= + \frac{4}{5}$



i) $\frac{\cot \frac{\theta}{2} + \tan \frac{\theta}{2}}{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}} = \sec \theta$

HS = $\frac{\frac{1}{t} + t}{\frac{1}{t} - t} \times t$
 $= \frac{1 + t^2}{1 - t^2}$

let $t = \tan \frac{\theta}{2}$
 $\therefore \frac{1}{t} = \cot \frac{\theta}{2}$

$= \frac{1}{\cos \theta}$ as $\cos \theta = \frac{1 - t^2}{1 + t^2}$
 $= \sec \theta$
 $= \text{HS}$

i) $\int_0^{1/4} \frac{dx}{1+16x^2} = \int_0^{1/4} \frac{dx}{16(\frac{1}{16}x^2)}$ (3)
 $= \frac{1}{16} \int_0^{1/4} \frac{\frac{1}{4}}{\frac{1}{16}x^2} dx$
 $= \frac{1}{16} \times 4 \left[\tan^{-1} \frac{x}{1/4} \right]_0^{1/4}$
 $= \frac{1}{4} \left[\tan^{-1} 4x \right]_0^{1/4}$
 $= \frac{1}{4} \left[\tan^{-1} 1 - \tan^{-1} 0 \right] = \frac{1}{4} \cdot \frac{\pi}{4} = \frac{\pi}{16}$

$\alpha + \beta + \gamma = 45^\circ$

$\tan \alpha = \frac{1}{2} \quad \tan \beta = \frac{1}{4}$

find $\tan \gamma$

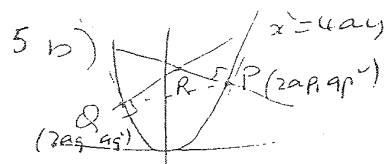
$\tan \gamma = \tan (45 - \beta - \alpha)$
 $= \tan (45 - (\alpha + \beta))$

now $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$
 $= \frac{\frac{1}{2} + \frac{1}{4}}{1 - \frac{1}{2} \cdot \frac{1}{4}}$
 $= \frac{\frac{3}{4}}{\frac{7}{8}}$
 $= \frac{3}{8} \times \frac{8}{7} = \frac{3}{7}$

$\therefore \tan (45 - (\alpha + \beta)) = \frac{\tan 45 - \tan(\alpha + \beta)}{1 + \tan 45 \cdot \tan(\alpha + \beta)}$
 $= \frac{1 - \frac{3}{7}}{1 + 1 \cdot \frac{3}{7}}$
 $= \frac{\frac{4}{7}}{\frac{10}{7}} = \frac{4}{10} = \frac{2}{5}$

$\tan (45 - (\alpha + \beta)) = \frac{2}{5}$
 $\tan \gamma = \frac{2}{5}$

4



$$x^2 = 4ay$$

$$\therefore y = \frac{1}{4a} x^2$$

$$y' = \frac{1}{2a} x$$

$$\text{when } x = 2ap \quad \text{slope of tangent} = \frac{2ap}{2a}$$

$$\therefore \text{slope of normal} = -\frac{1}{p}$$

$$\therefore \text{eqn of normal } y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$\therefore x + py = 2ap + ap^3$$

$$x + py = a(2p + p^3) \quad (*)$$

$$ii) \quad x + py = 2ap + ap^3 \quad \text{--- (1)}$$

$$x + qy = 2aq + aq^3 \quad \text{--- (2)}$$

$$+ \text{eqn (1) - (2)}$$

$$(p - q)y = 2a(p - q) + a(p^3 - q^3)$$

$$(p - q)y = 2a(p - q) + a(p - q)(p^2 + pq + q^2) \quad p \neq q$$

$$\therefore y = 2a + a(p^2 + pq + q^2) \quad \text{--- (3)}$$

$$\text{Sub (3) into (1)}$$

$$x + p(2a + a(p^2 + pq + q^2)) = 2ap + ap^3$$

$$x + 2ap + ap^3 + ap^2q + apq^2 = 2ap + ap^3$$

$$x = -apq(p + q)$$

$$\therefore R(-apq(p + q), 2a + a(p^2 + pq + q^2))$$

$$iii) \quad P(2ap, ap^2) \quad Q(2aq, aq^2)$$

$$\text{slope of } PQ = \frac{aq^2 - ap^2}{2aq - 2ap}$$

$$= \frac{a(q^2 - p^2)}{2a(q - p)} = \frac{a(q + p)}{2a}$$

$$= \frac{p + q}{2}$$

$$y - ap^2 = \frac{p + q}{2} (2a - 2ap)$$

$$y - ap^2 = \left(\frac{p + q}{2}\right)x - ap^2 - apq$$

$$y = \left(\frac{p + q}{2}\right)x - apq$$

$$(iv) \quad PQ: y = \left(\frac{p + q}{2}\right)x - apq$$

$$\text{passes through } (0, 2a)$$

$$\therefore 2a = \left(\frac{p + q}{2}\right) \cdot 0 - apq$$

$$pq = -2 \quad \text{--- (1)}$$

$$\text{Locus of } R:$$

$$x = -apq(p + q)$$

$$x = 2(p + q) \quad \text{from (1)}$$

$$p + q = \frac{x}{2a}$$

$$y = 2a + a(p^2 + pq + q^2)$$

$$y = 2a + a((p + q)^2 - pq)$$

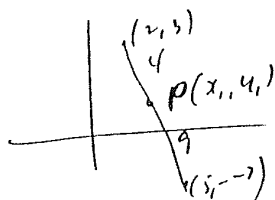
$$y = 2a + a\left(\left(\frac{x}{2a}\right)^2 + 2\right)$$

$$y = 2a + \frac{x^2}{4a} + 2a$$

$$y = 4a + \frac{x^2}{4a}$$

$$4ay = x^2 + 16a^2$$

6 a)



(2)

$$x_1 = \frac{4 \times 5 + 9 \times 2}{4 + 9} = 2 \frac{12}{13}$$

$$y_1 = \frac{4 \times 5 - 7 + 9 \times 3}{4 + 9} = -\frac{1}{13}$$

$$\therefore p(2 \frac{12}{13}, -\frac{1}{13})$$

6 b) (i) $\frac{ds}{dt} = 0.01$ (1)

$$s = 4\pi r^2$$

$$\frac{ds}{dt} = \frac{ds}{dr} \cdot \frac{dr}{dt}$$

$$\frac{ds}{dt} = 8\pi r \cdot \frac{dr}{dt}$$

when $r = 5$

$$0.01 = 8\pi \cdot 5 \cdot \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{0.01}{40\pi}$$

$$= 7.96 \times 10^{-5}$$

rate of change of radius is

(ii)

$$V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$= 4\pi r^2 \cdot \frac{dr}{dt}$$

when $r = 5$

$$\frac{dV}{dt} = 4\pi (5^2) \cdot 7.96 \times 10^{-5} = 0.025$$

6 c) $\frac{d}{dx} \sin^{-1}(e^{2x}) = \frac{1}{\sqrt{1-e^{4x}}} \cdot 2e^{2x}$

(2)

$$\int_{-\ln \sqrt{2}}^0 \frac{2e^{2x}}{\sqrt{1-e^{4x}}} dx = \left[\sin^{-1}(e^{2x}) \right]_{-\ln \sqrt{2}}^0$$

$$= \sin^{-1}(e^0) - \sin^{-1}(e^{2(-\ln \sqrt{2})})$$

$$= \sin^{-1}(1) - \sin^{-1}(e^{-2 \ln \sqrt{2}})$$

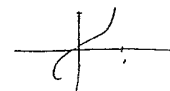
$$= \sin^{-1}(1) - \sin^{-1}(e^{\ln(2)^{-2}})$$

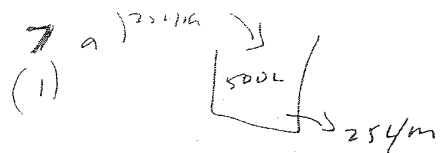
$$= \frac{\pi}{2} - \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{2} - \frac{\pi}{6}$$

$$= \frac{\pi}{3}$$

(3)





Amount of salt
in tank
initially is 0

$$t=0$$

$$A=0$$

Brine going in:

Concentration of salt = $\frac{1}{10}$ solute

Rate of salt going in = $25 \times \frac{1}{10}$ 4/mi (1)

$$= \frac{25}{10}$$

$$= 2.5 \text{ 4/mi}$$

Running out:

Concentration of salt = $\frac{A}{500} \times 25$ 4/min (1)

$$= \frac{25A}{500}$$

$$= \frac{A}{20} \text{ 4/min}$$

$$\frac{dA}{dt} = 2.5 - \frac{A}{20}$$

$$= \frac{50 - A}{20}$$

$$\frac{dA}{dt} = -\frac{1}{20}(A - 50)$$

(11) if $\frac{dA}{dt} = k(A - b)$ (1)

$$A = B + Ae^{kt}$$

So $A = 50 + A_0 e^{-\frac{1}{20}t}$

when $T=0$
 $A=0$

$$0 = 50 + A_0 e^0$$

$$A_0 = -50$$

$$A = 50 - 50 e^{-\frac{1}{20}t}$$

at 100 min $H = 50 - 50 e^{-\frac{1}{20} \times 100}$ (1)

$$= 50(1 - e^{-5})$$

$$= 49.6669$$

V/

$$x = mx + b$$

$$\frac{d}{dt}(\frac{1}{2}v^2) = mx + b$$

$$t=0$$

$$x=0$$

$$v=0$$

$$a=3$$

$$\frac{1}{2}v^2 = \frac{m}{2}x^2 + bx + c$$

$$v^2 = mx^2 + 2bx + c$$

$$0 = 0 + c$$

$$v^2 = mx^2 + 2bx$$

max speed
 $x=160$

now $\ddot{x} = mx + b$

$$a=3$$

$$x=0$$

$$3 = 0 + b$$

$$\boxed{\ddot{x} = mx + 3}$$

It reaches max speed when $x=160$

ii $\frac{dv}{dt} = 0$

$$\therefore \ddot{x} = 0 \text{ when } x=160$$

$$0 = 160m + 3$$

$$m = -\frac{3}{160}$$

$$v^2 = -\frac{3}{160}x^2 + 6x$$

(1) max speed when $x=160$

$$v^2 = -\frac{3}{160} \cdot 160^2 + 6 \cdot 160$$

$$= 480$$

$$v = \sqrt{480}$$

$$= \pm 4\sqrt{30} \text{ m/s}$$

max speed is $4\sqrt{30} \text{ m/s}$

(11) speed when $x=80$

$$v^2 = -\frac{3}{160}(80)^2 + 6 \cdot 80$$

$$= 360$$

$$v = \pm \sqrt{360} = \pm 6\sqrt{10}$$

$\therefore$ speed is $6\sqrt{10} \text{ m/s}$